PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON EXPERIENCE (IOHE)

On

Smart Enterprise Employee Management System

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING AND TECHNOLOGY

CHITKARA UNIVERSITY, PUNJAB, INDIA

DECLARATION

I hereby certify that the work which is being presented in the project report entitled **Smart Enterprise (Employee Management System)** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of **Dr. Rajat Takkar**. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura
Date: 28-04-2026

Vivek Singh
University Roll No: 2211981468

This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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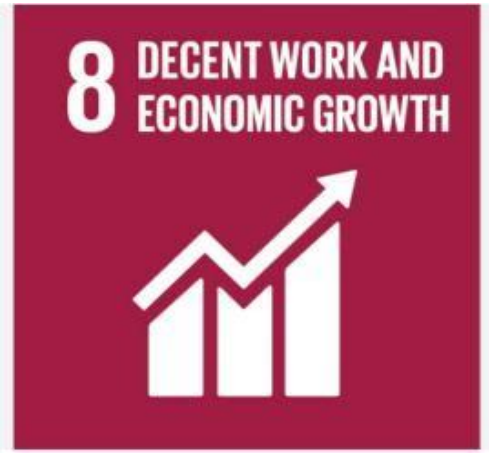
SDG Goals

1: SDG - 8



2. SDG - 9

ABSTRACT

TaskMate is a comprehensive digital task management platform designed to facilitate seamless assignment, tracking, and completion of work tasks within an organization through a secure, transparent, and user-friendly online environment. The platform aims to simplify traditional task management processes by integrating essential features that enhance accessibility, trust, and operational efficiency for both administrators and employees.

The core functionality of TaskMate enables administrators to create and assign tasks to employees, while allowing employees to manage their workloads efficiently through a structured and organized system. One of the key highlights of the platform is that users can view and interact with tasks without complex navigation, encouraging wider engagement by allowing employees to explore assigned tasks, update statuses, and review detailed specifications before completion.

To improve transparency and team confidence, TaskMate provides full task details, including titles, descriptions, due dates, categories, and attachments, ensuring that users can carefully assess task requirements visually and functionally before taking action. Additionally, the platform incorporates a direct status update mechanism that allows employees to mark tasks as active, completed, or failed in real time. This built-in update system eliminates manual tracking and fosters secure and clear progress reporting between administrators and employees.

TaskMate also integrates a secure data persistence mechanism to ensure safe and protected task and user information storage. The system is designed with local storage and state management protocols to safeguard data and maintain continuity across sessions. Upon task completion or updates, the platform automatically reflects changes in dashboards and counters, providing users with an easily accessible and verifiable record of progress.

By combining open task browsing, real-time status updates, secure data handling, comprehensive task visualization, and automated progress tracking, TaskMate establishes a modern and reliable digital task management ecosystem. The platform enhances organizational trust, simplifies task workflows, and ensures operational transparency, making it a scalable and innovative solution within the online workforce management sector.

1.1 Problem Background

In today's fast-paced business environment, effective task management is crucial for organizational success. However, many companies still rely on traditional, manual methods for assigning, tracking, and completing work tasks, leading to significant inefficiencies and operational bottlenecks. These outdated approaches often result in poor communication, lack of transparency, and reduced productivity, ultimately impacting employee satisfaction and business outcomes.

1.2 Objective of the Project

The primary objective of the **TaskMate** project is to develop a comprehensive, web-based platform that addresses the inefficiencies of traditional task management in organizations. The system aims to streamline the processes of task assignment, tracking, and completion by providing a centralized, user-friendly interface that enhances communication, transparency, and productivity.

To achieve this overarching goal, the following specific objectives have been defined:

1. Role-Based Access and Security : To design and implement a secure authentication system that distinguishes between administrative and employee-level users, ensuring that each role has access only to the features and data relevant to their responsibilities. This prevents unauthorized access and maintains data integrity across the platform.

2. Efficient Task Assignment and Delegation : To provide administrators with a powerful, intuitive dashboard that enables them to create, assign, and distribute tasks to specific employees with clearly defined priorities, deadlines, and descriptions — eliminating the ambiguity and delays associated with verbal or email-based task delegation.

3. Real-Time Task Tracking and Visibility : To enable both administrators and employees to monitor the status of tasks in real time, offering complete visibility into what has been completed, what is in progress, and what is pending — thereby reducing bottlenecks and improving accountability across teams.

4. Employee Empowerment and Engagement : To equip employees with a personalized dashboard that presents their assigned tasks clearly, allows them to update task statuses independently, and keeps them informed of new assignments through an integrated notification system — fostering a sense of ownership and responsibility.

5. Seamless User Experience : To deliver a clean, responsive, and accessible interface built with modern frontend technologies, featuring dark mode support and intuitive navigation, so that users across varying technical backgrounds can adopt and use the platform with minimal training.

1.3 Proposed Solution / System Overview

The proposed solution is a full-stack web application designed as a digital task management platform. The system operates on a client-side architecture with local data persistence, eliminating the need for a backend server for core functionality. TaskMate provides two primary user roles: administrators and employees, each with tailored dashboards and capabilities.

Administrators can create tasks with detailed specifications, assign them to employees via a dropdown selection, and monitor overall team progress through summary views. Employees access a personalized task list where they can view, filter, search, and update task statuses.

1.4 Key Features or Functionality

TaskMate incorporates a suite of features designed to optimize task management workflows across all levels of an organization. Each feature has been thoughtfully engineered to address a specific pain point in traditional task coordination, delivering a seamless and productive experience for both administrators and employees.

1. User Authentication and Role-Based Access Control : TaskMate implements a secure login and sign-up system that differentiates between two distinct user roles — Admin and Employee. Each role is granted access exclusively to the features and data relevant to their position within the organization. Administrators are provided with elevated privileges to manage the entire workforce and task pipeline, while employees are presented with a focused, personalized view of their own assignments. Session persistence ensures that authenticated users remain logged in across page reloads, eliminating the friction of repeated sign-ins and maintaining workflow continuity.

2. Task Creation and Assignment The administrative interface provides a comprehensive task creation module that allows admins to define tasks with a rich set of attributes, including a task title, detailed description, due date, category classification, and file attachments. Tasks are assigned to specific employees through an intuitive dropdown selector, ensuring that responsibilities are clearly delegated with no ambiguity. This structured approach to task creation replaces ad hoc communication methods and establishes a single source of truth for all pending work within the organization.

3. Task Status Management Employees are empowered to actively manage the lifecycle of their assigned tasks through a clearly defined set of status transitions. Upon receiving a new task, an employee may accept it to move it into an active state, subsequently marking it as completed upon finishing or as failed if the task could not be fulfilled within the given

constraints. Each status change triggers automatic counter updates across the dashboard, providing administrators with an always-current overview of team productivity and task progress without the need for manual reporting.

4. Advanced Filtering, Search, and Sorting To help users navigate large volumes of tasks efficiently, TaskMate offers a powerful set of filtering, search, and sorting capabilities. Tasks can be filtered by status categories — including All, New, Active, Completed, Failed, and Due Soon — allowing users to instantly isolate the subset of tasks most relevant to their current focus. A keyword-based search enables users to locate specific tasks by title in real time, while sorting controls allow tasks to be ordered by due date in either ascending or descending order. Together, these tools ensure that no task is ever buried or overlooked within a growing task list.

5. File Attachments TaskMate supports the uploading of multiple file types alongside task records, including images, PDF documents, and plain text files. All uploaded files undergo client-side validation to ensure compatibility and integrity before being encoded in base64 format for storage and display. Attached files are visually indicated on their respective task cards, giving both administrators and employees immediate awareness of supporting materials without needing to open each task individually. This feature reduces the need for separate file-sharing tools by consolidating task content and its associated resources in one place.

6. **Due Date Notifications and Urgency Indicators** To prevent missed deadlines and promote timely task completion, TaskMate incorporates a proactive due date notification system. Tasks that have passed their due date are flagged with a prominent overdue badge, while tasks approaching their deadline within a defined threshold are marked with a due-soon indicator. A dedicated filter for urgent items allows employees and administrators to surface time-sensitive tasks instantly, enabling prioritization decisions to be made quickly and confidently. This system acts as a continuous, passive reminder that keeps all stakeholders aligned with the organization's timelines and commitments.

7. Dark Mode and User Interface Personalization TaskMate offers a built-in dark mode toggle powered by a globally accessible context provider, allowing users to switch between light and dark themes according to their personal preference or ambient environment. This setting is applied consistently across all components and dashboards in real time, without requiring a page reload. By prioritizing visual comfort and accessibility, TaskMate ensures that users can engage with the platform for extended periods without eye strain — a consideration that is particularly valuable for employees who rely on the system as a core part of their daily workflow. The clean, responsive interface built with Tailwind CSS further ensures a consistent and polished experience across devices of varying screen sizes.

1.5 Technology / Method Used

- Frontend Framework: React 18 with functional components and hooks (useState, useContext, useEffect) for efficient state management and component lifecycle handling.

- Build Tool and Development Server: Vite for fast development, hot module replacement, and optimized production builds.

- Styling Framework: Tailwind CSS for utility-first, responsive styling that ensures mobile compatibility and consistent design.

- State Management: React Context API for global state sharing across components, enabling real-time updates without prop drilling.

- Data Persistence: Browser localStorage for client-side data storage, with JSON serialization for complex objects and automatic persistence on state changes.

1.6 Expected Outcome & Benefits

The successful implementation of TaskMate is expected to deliver significant improvements in organizational task management

- Enhanced Productivity: Streamlined task assignment and tracking reduce manual efforts, allowing teams to focus on core work rather than administrative tasks.

- Improved Transparency and Accountability: Real-time status updates and visual indicators provide clear visibility into task progress, fostering better communication and responsibility.

- Reduced Errors and Delays: Automated persistence and validation minimize data loss and inconsistencies, while due date notifications help prevent missed deadlines.

1.7 Original Contribution Statement

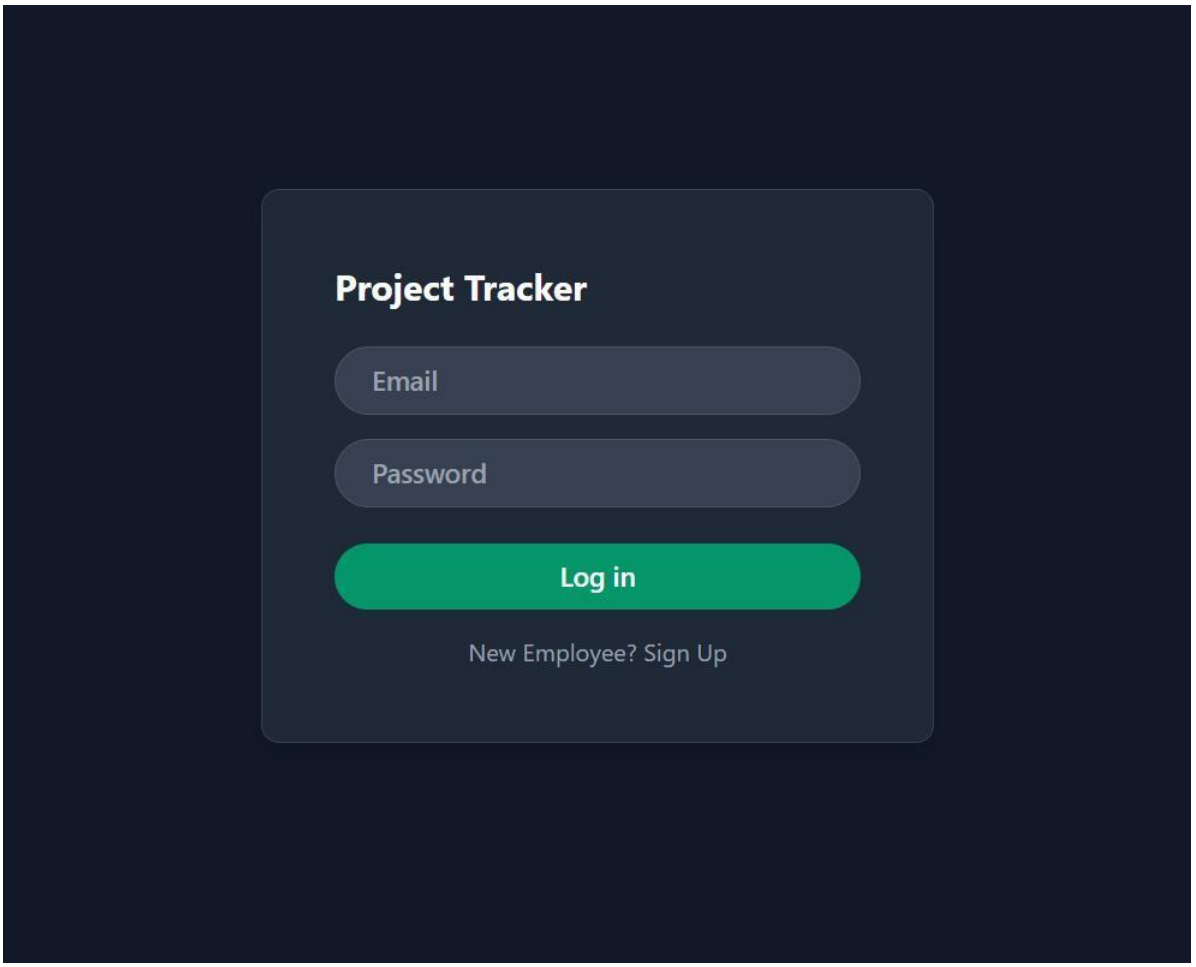
This project represents an original contribution to the field of web-based task management systems by developing a comprehensive, client-side solution that integrates advanced features typically found in enterprise-grade applications. Through the deliberate and innovative combination of modern frontend technologies, user-centric design principles, and practical organizational functionality, TaskMate establishes a meaningful addition to the landscape of open-source productivity tools.

1. Novel Integration of Enterprise Features in a Lightweight Architecture TaskMate bridges the gap between heavyweight enterprise task management platforms and overly simplistic to-do applications by delivering features such as file attachments with base64 encoding, role-based access control, real-time status tracking, and due date urgency indicators — all within a purely client-side React architecture. This combination, which is rarely found in open-source task managers of comparable scope, demonstrates that enterprise-level functionality need not depend on complex backend infrastructures or costly third-party services to be both effective and maintainable.

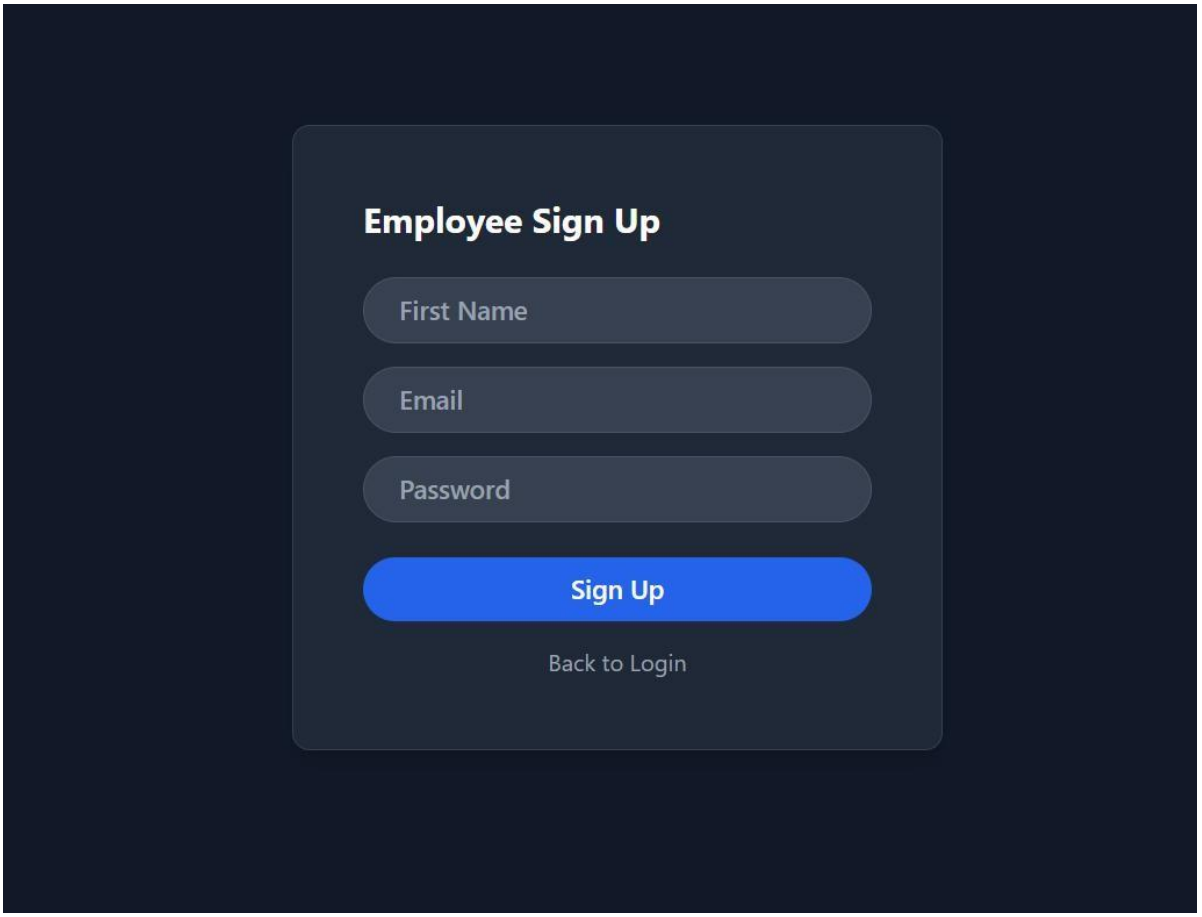
2. Role-Based Architecture as a Design Contribution The implementation of a dual-role system — separating administrative and employee concerns at both the interface and data access levels — contributes a practical and replicable pattern for role-based frontend architecture using React's Context API. Rather than relying on server-side role enforcement, TaskMate demonstrates how access control, conditional rendering, and state isolation can be achieved cleanly and securely on the client side, offering a valuable reference model for developers building similar systems without dedicated backend infrastructure.

3. **Unified Task Lifecycle Management** A distinctive contribution of TaskMate lies in its holistic treatment of the task lifecycle — from creation and assignment through acceptance, active progression, completion, or failure — within a single, cohesive interface. Unlike many existing tools that treat task creation and tracking as separate concerns, TaskMate unifies these processes under one platform, with automatic counter synchronization and state transitions that reflect real-world organizational workflows. This end-to-end lifecycle model offers new insights into how task state machines can be implemented effectively in a React component-based environment.

Login Page:



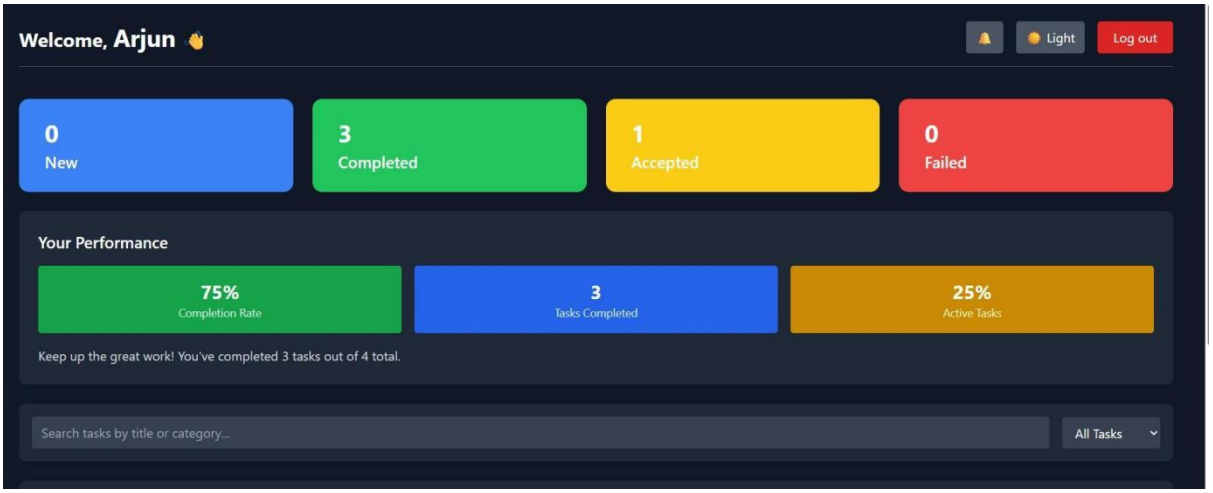
Signup Page:

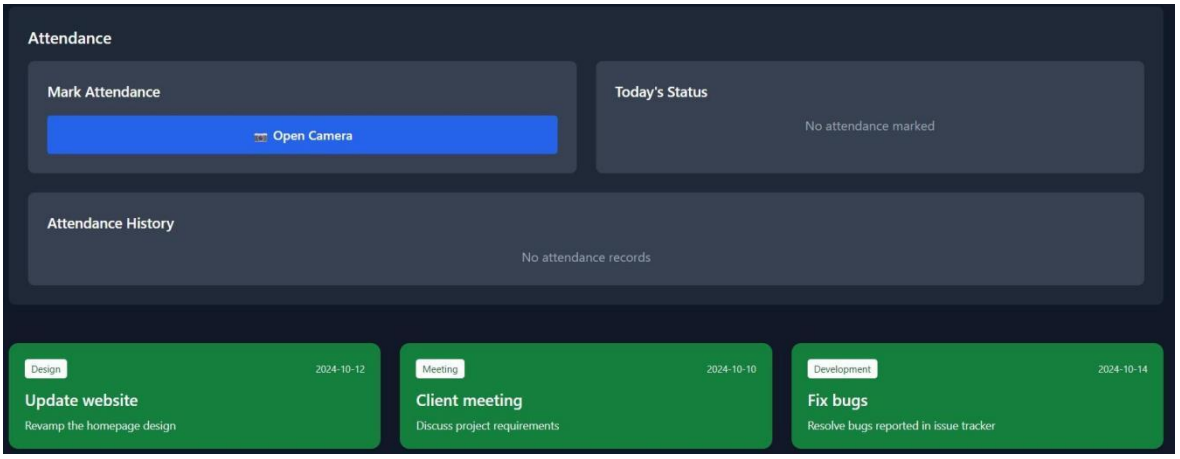

Employee Sign Up
First Name
Email
Password
Sign Up
Back to Login

```
ems-main > src > App.jsx > App > handleLogin
const App = () => {
  useEffect(()=>{
    const loggedInUser = localStorage.getItem('loggedInUser')

    if(loggedInUser){
      const userData = JSON.parse(loggedInUser)
      setUser(userData.role)
      setLoggedInUserData(userData.data)
    }

  },[])


  const handleLogin = (email, password) => {
    if (email === 'vivek@me.com' && password === '6969') {
      setUser('admin')
      localStorage.setItem('loggedInUser', JSON.stringify({ role: 'admin' }))
      return
    }

    if (!userData) {
      alert('Please wait a moment and try again.')
      return
    }

    const employee = userData.find((e) => email === e.email && password === e.password)
    if (employee) {
      setUser('employee')
      setLoggedInUserData(employee)
      localStorage.setItem('loggedInUser', JSON.stringify({ role: 'employee', data: employee }))
      return
    }

    alert('Invalid credentials')
  }

  const handleSignup = (firstName, email, password) => {
    if (!userData) {
      alert('Please wait a moment and try again.')
```

```
ems-main > src > App.jsx > App > handleLogin
const App = () => {
  const handleLogin = (email, password) => {
  }

  const handleSignup = (firstName, email, password) => {
    if (!userData) {
      alert('Please wait a moment and try again.')
      return
    }

    const exists = userData.find((u) => u.email === email)
    if (exists) {
      alert('Email is already registered')
      return
    }

    const nextId = userData.reduce((max, u) => Math.max(max, u.id || 0), 0) + 1

    const newUser = {
      id: nextId,
      firstName,
      email,
      password,
      taskCounts: { active: 0, newTask: 0, completed: 0, failed: 0 },
      tasks: []
    }

    const updatedEmployees = [...userData, newUser]
    setUserData(updatedEmployees)
    localStorage.setItem('employees', JSON.stringify(updatedEmployees))

    setUser('employee')
    setLoggedInUserData(newUser)
    localStorage.setItem('loggedInUser', JSON.stringify({ role: 'employee', data: newUser }))
    setShowSignup(false)
  }


  return (
```

Front Page :


Welcome, Arjun
Light
Log out
0
New
3
Completed
1
Accepted
0
Failed
Your Performance
75%
Completion Rate
3
Tasks Completed
25%
Active Tasks
Keep up the great work! You've completed 3 tasks out of 4 total.
Search tasks by title or category...
All Tasks


Attendance
Mark Attendance
Open Camera
Today's Status
No attendance marked
Attendance History
No attendance records
Design
2024-10-12
Update website
Revamp the homepage design
Meeting
2024-10-10
Client meeting
Discuss project requirements
Development
2024-10-14
Fix bugs
Resolve bugs reported in issue tracker

File Edit Selection View Go Run ···

Vivek-Project

Attendance.jsx X

ems > src > components > other > Attendance.jsx > ...

```jsx
import React, { useState, useRef, useContext } from 'react'
import { AuthContext } from '../../context/AuthProvider'

const Attendance = ({ employeeData }) => {
    const [userData, setUserData] = useContext(AuthContext)
    const [useCamera, setUseCamera] = useState(false)
    const [isCheckingIn, setIsCheckingIn] = useState(false)
    const videoRef = useRef(null)
    const canvasRef = useRef(null)
    const [successMessage, setSuccessMessage] = useState('')

    const todayAttendance = employeeData?.attendance?.find(att => {
        const attDate = new Date(att.date)
        const today = new Date()
        return attDate.toDateString() === today.toDateString()
    })

    const startCamera = async () => {
        try {
            const stream = await navigator.mediaDevices.getUserMedia({
                video: {
                    facingMode: 'user',
                    width: { ideal: 320 },
                    height: { ideal: 240 }
                },
                audio: false
            })
            if (videoRef.current) {
                videoRef.current.srcObject = stream
                videoRef.current.play()
            }
            setUseCamera(true)
        } catch (error) {
            alert('Unable to access camera. ' + error.message)
        }
    }

```

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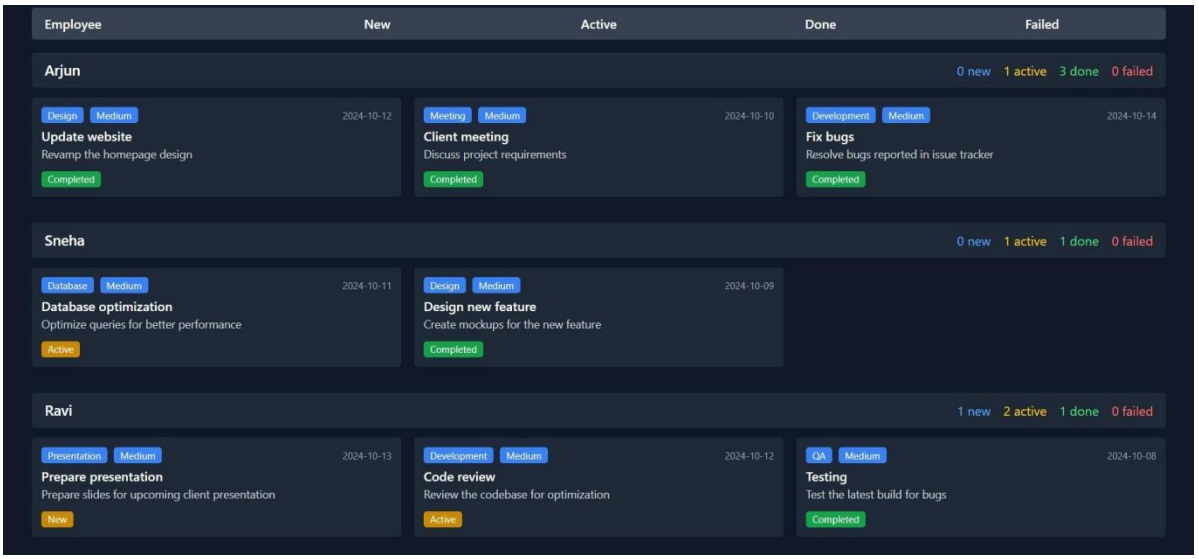
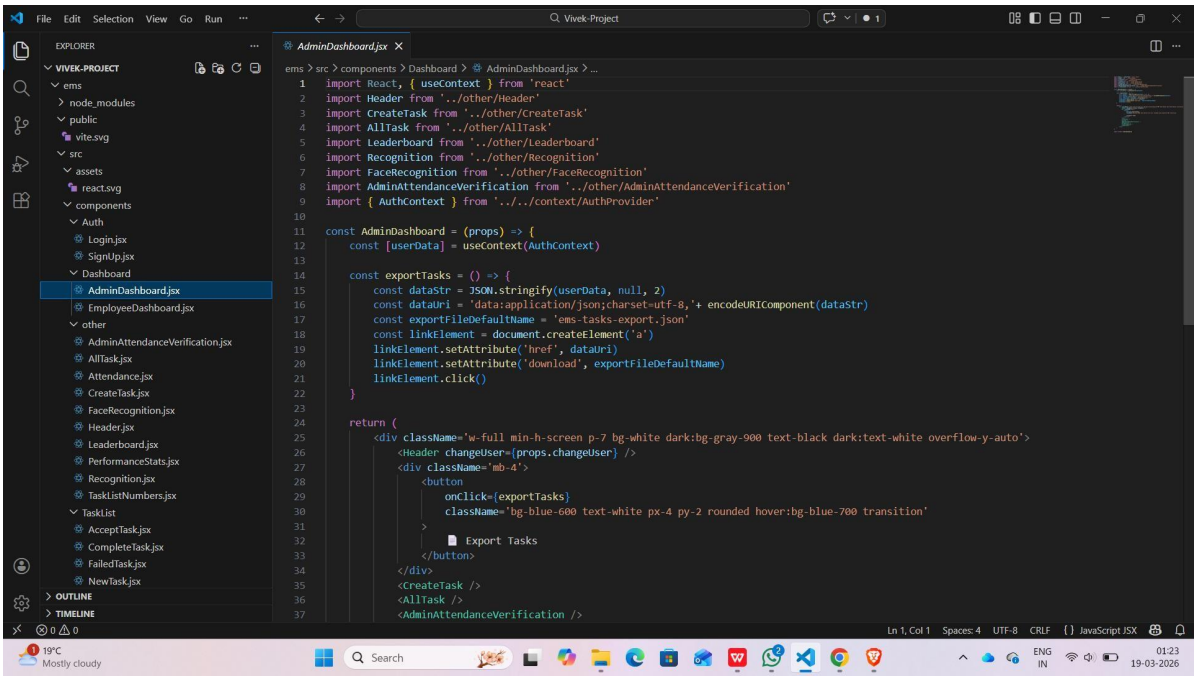
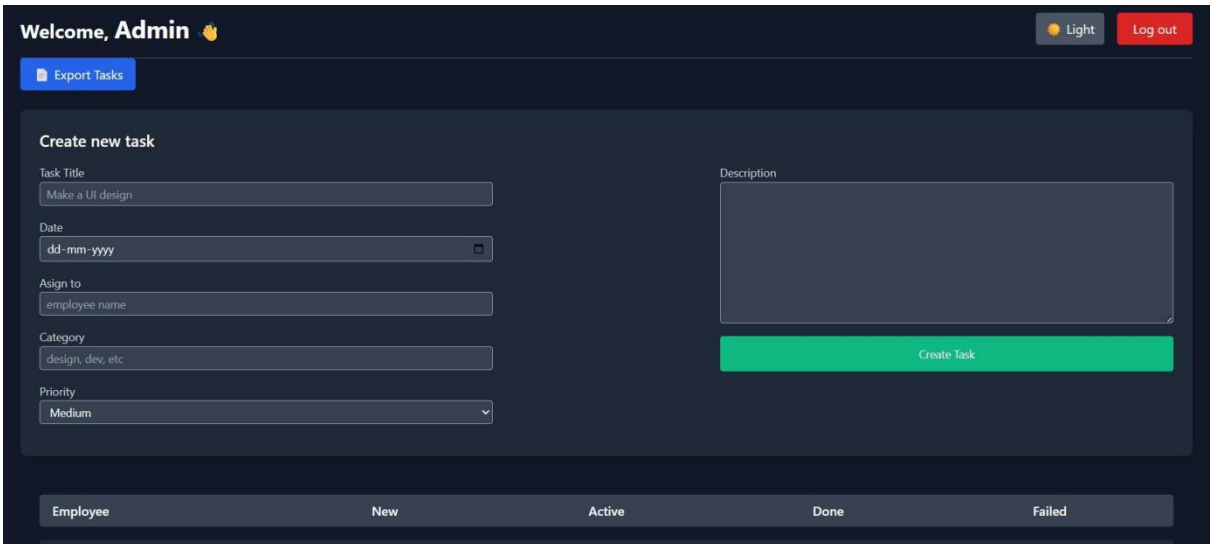
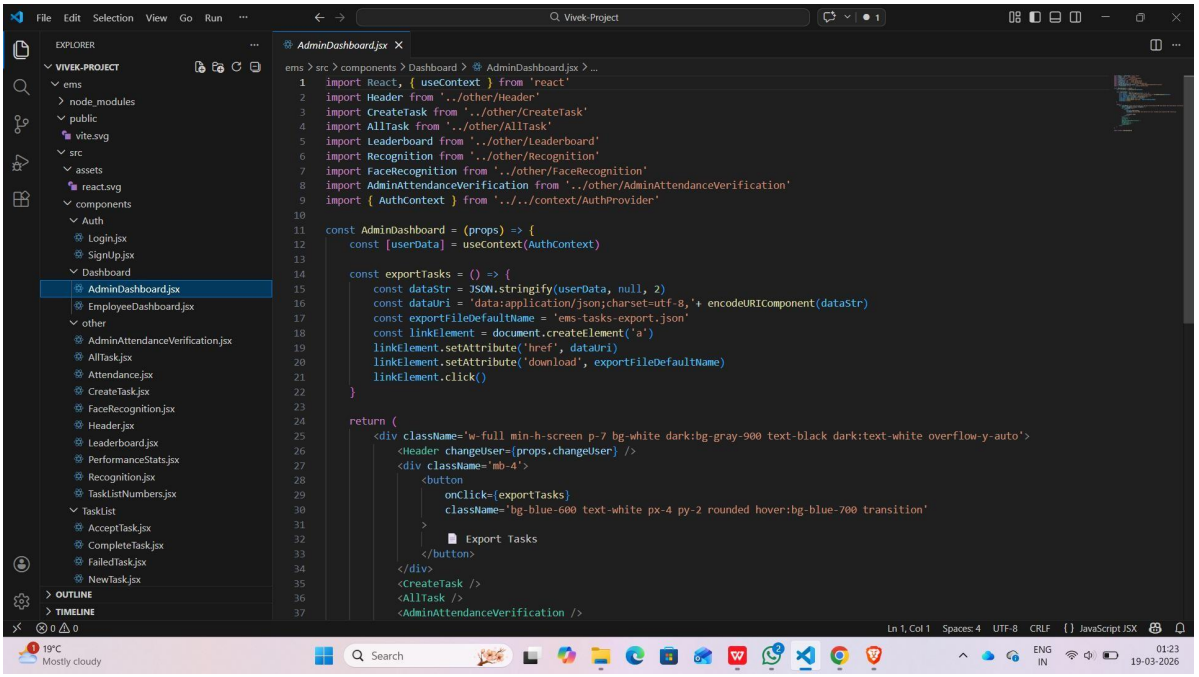
Employee Task Dashboard :





Admin Page :


Welcome, Admin
Light
Log out
Export Tasks
Create new task
Task Title
Make a UI design
Date
dd-mm-yyyy
Asign to
employee name
Category
design, dev, etc
Priority
Medium
Description
Create Task
Employee
New
Active
Done
Failed

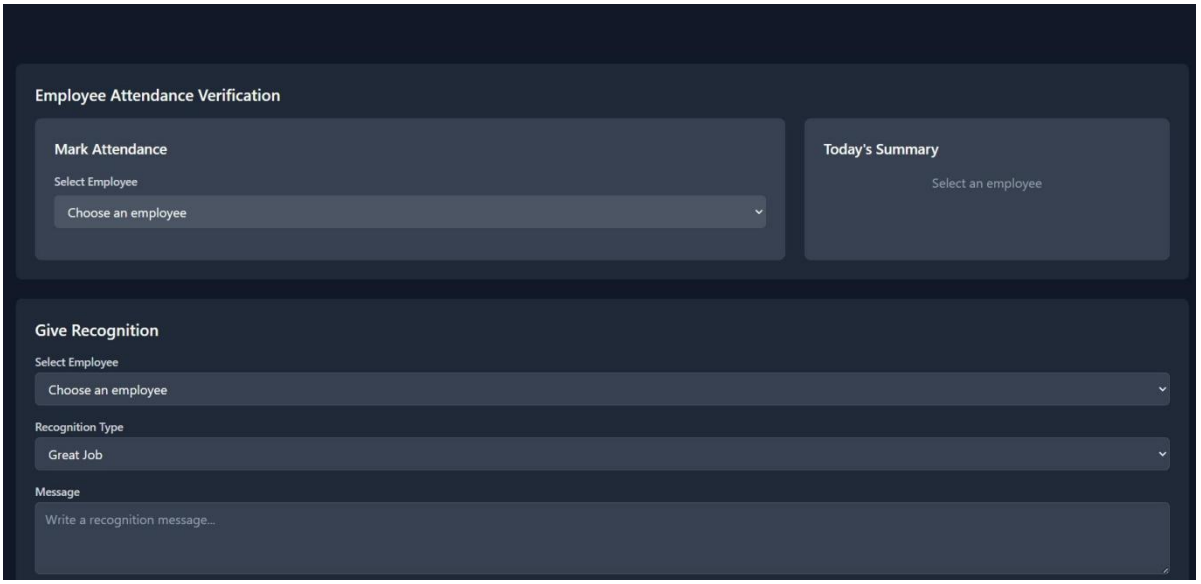

File Edit Selection View Go Run
Vivek-Project
EXPLORER
VIVEK-PROJECT
ems
node_modules
public
vite.svg
src
assets
react.svg
components
Auth
Login.jsx
SignUp.jsx
Dashboard
AdminDashboard.jsx
EmployeeDashboard.jsx
other
AdminAttendanceVerification.jsx
AllTask.jsx
Attendance.jsx
CreateTask.jsx
FaceRecognition.jsx
Header.jsx
Leaderboard.jsx
PerformanceStats.jsx
Recognition.jsx
TaskListNumbers.jsx
TaskList
AcceptTask.jsx
CompleteTask.jsx
FailedTask.jsx
NewTask.jsx
OUTLINE
TIMELINE
AdminDashboard.jsx
ems > src > components > Dashboard > AdminDashboard.jsx > ...
import React, { useContext } from 'react'
import Header from '../other/Header'
import CreateTask from '../other/CreateTask'
import AllTask from '../other/AllTask'
import Leaderboard from '../other/Leaderboard'
import Recognition from '../other/Recognition'
import FaceRecognition from '../other/FaceRecognition'
import AdminAttendanceVerification from '../other/AdminAttendanceVerification'
import { AuthContext } from '../../context/AuthProvider'
const AdminDashboard = (props) => {
const [userData] = useContext(AuthContext)
const exportTasks = () => {
const dataStr = JSON.stringify(userData, null, 2)
const dataUri = 'data:application/json;charset=utf-8,'+ encodeURIComponent(dataStr)
const exportFileDefaultName = 'ems-tasks-export.json'
const linkElement = document.createElement('a')
linkElement.setAttribute('href', dataUri)
linkElement.setAttribute('download', exportFileDefaultName)
linkElement.click()
}
return (

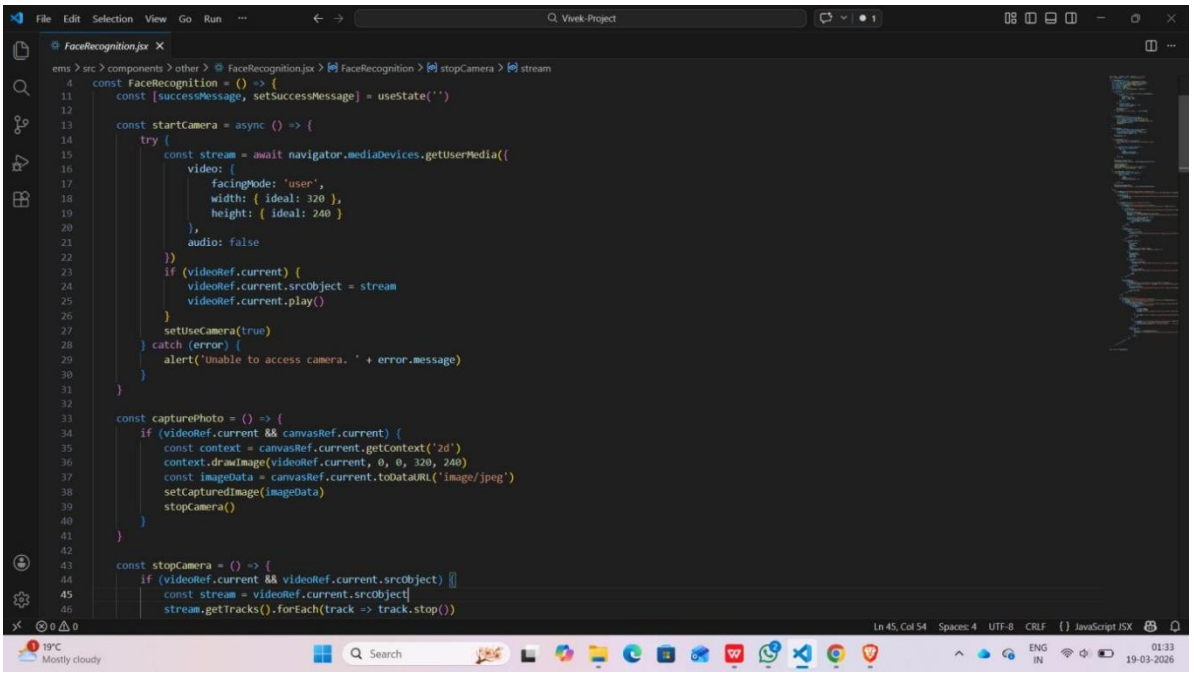
<Header changeUser={props.changeUser} />

<button
onClick={exportTasks}
className='bg-blue-600 text-white px-4 py-2 rounded hover:bg-blue-700 transition'
>
Export Tasks
</button>

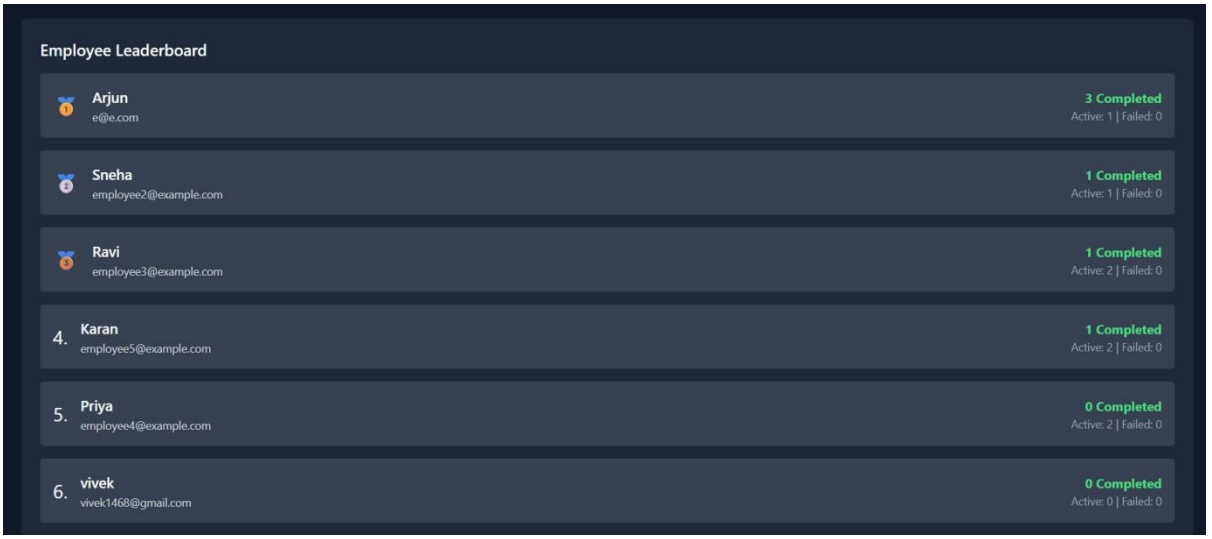
<CreateTask />
<AllTask />
<AdminAttendanceVerification />
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
JavaScript JSX
19°C
Mostly cloudy
Search
ENG
IN
01:23
19-03-2026

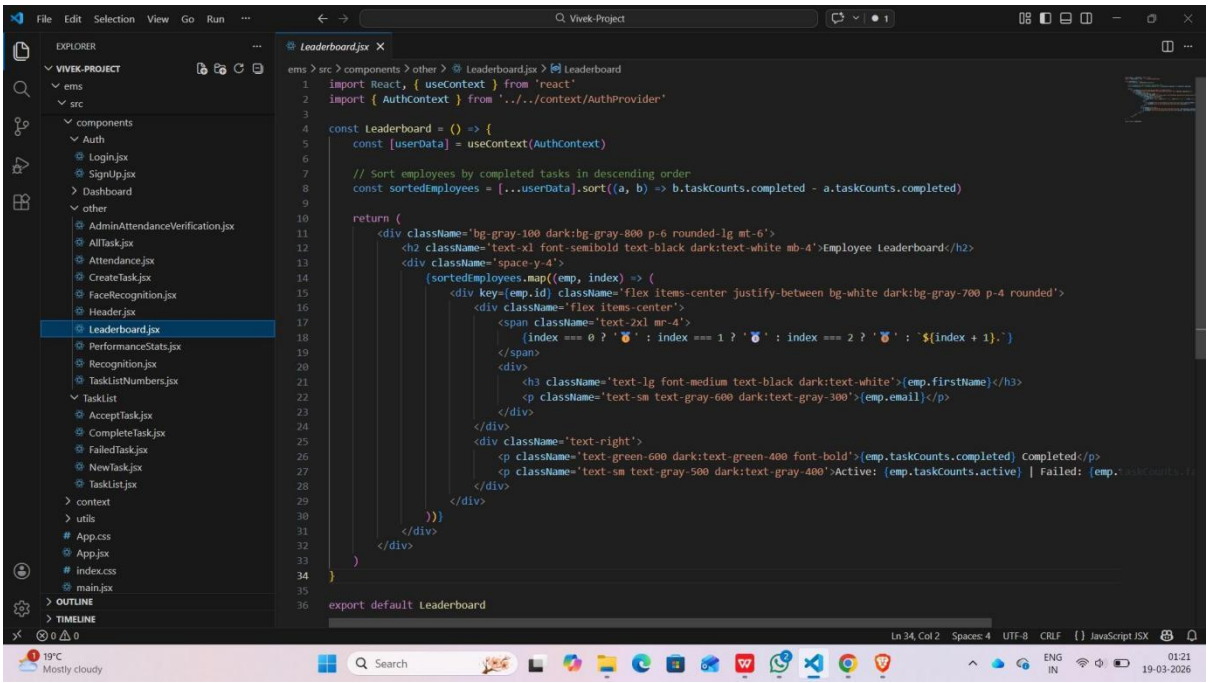
Attendance & recognition Dashboard :



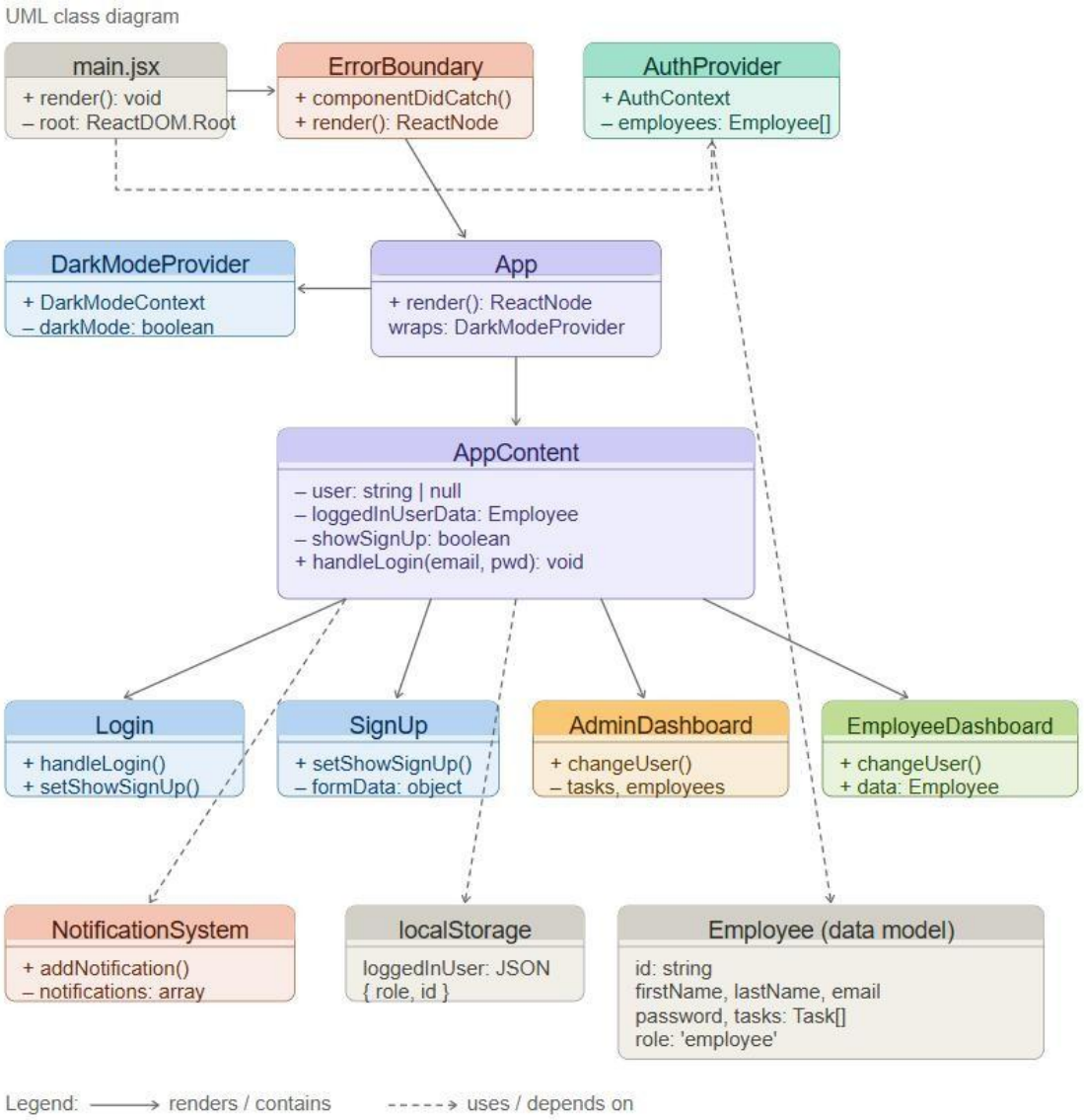


Employe Leaderboard Page :

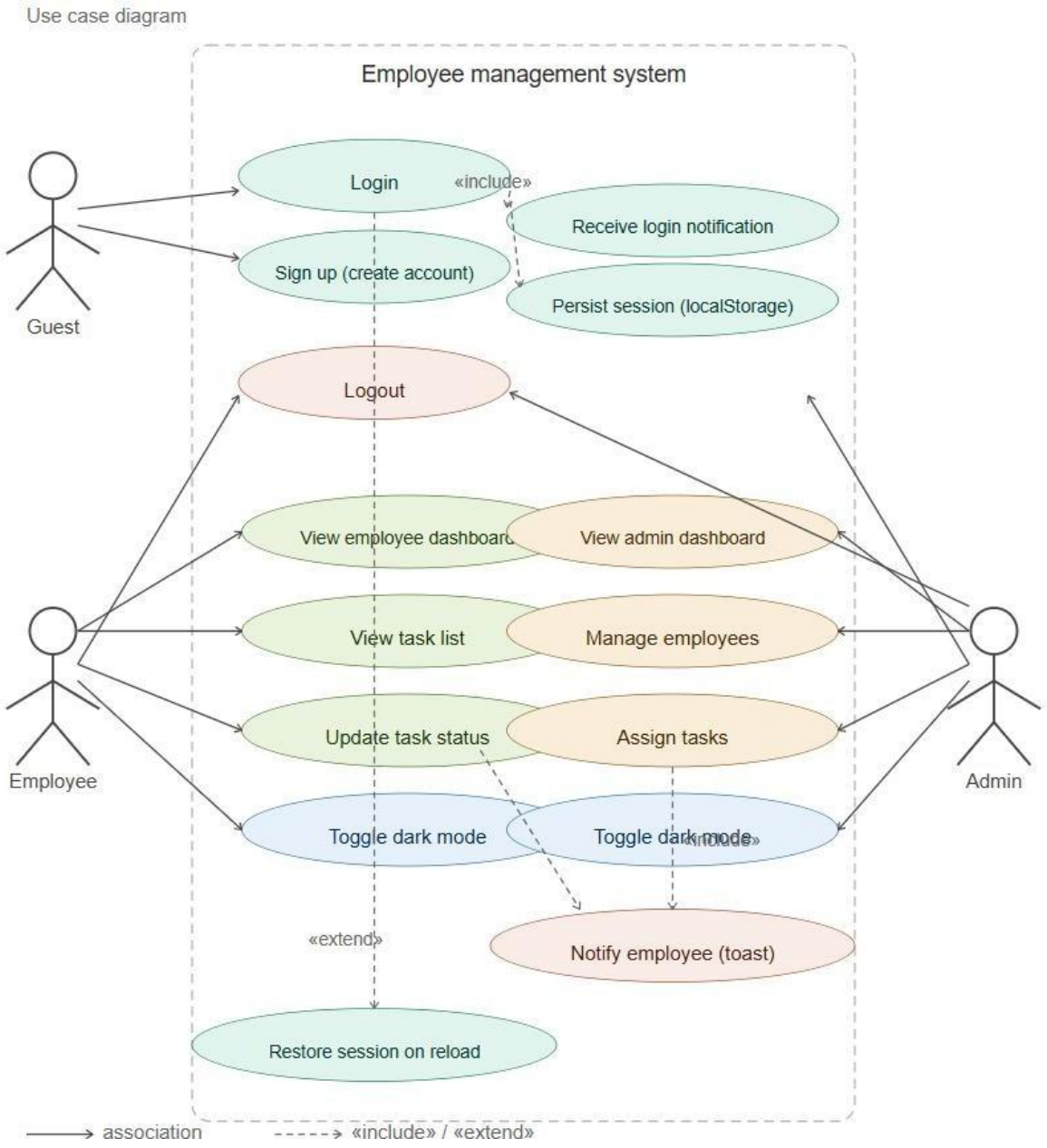




UML CLASS DIAGRAM



USECASE DIAGRAM



CONCLUSION :

The TaskMate system successfully demonstrates the development of a modern and efficient web-based task management platform that overcomes the limitations of traditional manual systems. By integrating features such as task creation, assignment, real-time status updates, and advanced filtering, the system ensures smooth and organized workflow management within an organization. The use of role-based access for administrators and employees enhances both security and usability.

The platform improves transparency by allowing users to clearly view task details, deadlines, and progress, which leads to better accountability and communication. Additionally, the implementation of local storage and state management ensures reliable data persistence and system performance without the need for a complex backend.

TaskMate also simplifies task tracking through an intuitive user interface built using modern technologies like React, Vite, and Tailwind CSS. Overall, the system increases productivity, reduces manual effort, and provides a centralized solution for managing tasks efficiently.
This project highlights the importance of digital transformation in organizational processes and serves as a strong foundation for future enhancements such as cloud integration and mobile support.